



ASTRAQUANTUM TECH

SUMMER OF CYBERSECURITY 2026

WEEK 8 PRACTICAL ASSIGNMENT

Digital Forensics — Disk Image Investigation with Autopsy

Evidence • Artifact Analysis • Deleted Files • Timeline Thinking

Student Name: Mariam Fatima

Student ID: AQT-1372

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1. Case Overview

This practical assignment involved the forensic investigation of a publicly available training disk image using Autopsy. The purpose of the investigation was to practice basic digital forensic procedures, including forensic case creation, evidence image analysis, artifact identification, deleted-file analysis, metadata examination, recent activity analysis, and keyword searching.

The investigation was conducted using a publicly available training image from the Digital Corpora NPS Test Disk Images dataset. The evidence was analyzed using Autopsy 4.23.1 without modifying the original forensic image.

The selected case was created specifically for the AstraQuantum Tech Summer of Cybersecurity 2026 Week 8 Digital Forensics Practical Assignment.

Case Summary

I analyzed the ntfs1-gen2.E01 forensic disk image using Autopsy. The investigation identified metadata associated with multiple PDF files and a keyword hit for the email address bullock@rti.org within Report02-3.pdf. These artifacts indicate that the corresponding information was present within the forensic image. However, the artifacts alone do not prove who created, accessed, or communicated the information.

2. Evidence / Forensic Image

The evidence used for this investigation was the publicly available ntfs1-gen2.E01 forensic disk image from the Digital Corpora NPS Test Disk Images collection.

Evidence Details

Item	Details
Evidence Image	ntfs1-gen2.E01
Evidence Type	Forensic Disk Image
Source	Digital Corpora – NPS Test Disk Images
Tool	Autopsy 4.23.1
Time Zone	Asia/Karachi (GMT+5:00)
Case Name	Week8_DigitalForensics_AQT-1372
Case Number	Week8_DigitalForensics_AQT-1372

The evidence image was stored separately from the Autopsy case directory. The original training image was not intentionally modified during the investigation.

Source: Digital Corpora – NPS Test Disk Images.

3. Autopsy Case Creation

A single-user forensic case was created in Autopsy 4.23.1 using the following case information:

Case Name: Week8_DigitalForensics_AQT-1372

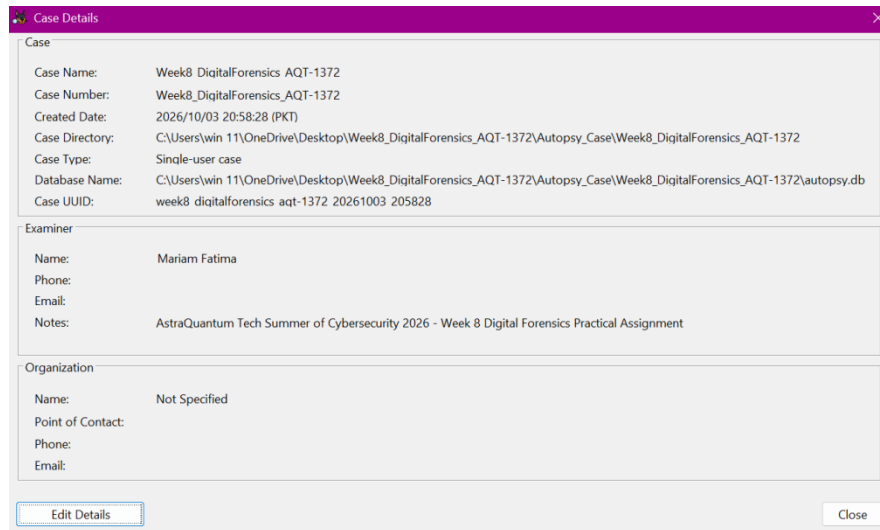
Case Number: Week8_DigitalForensics_AQT-1372

Examiner Name: Mariam Fatima

Case Type: Single-user case

Case Notes: AstraQuantum Tech Summer of Cybersecurity 2026 – Week 8 Digital Forensics Practical Assignment

The case was stored separately from the evidence image to maintain an organized forensic working environment.



Case	
Case Name:	Week8_DigitalForensics_AQT-1372
Case Number:	Week8_DigitalForensics_AQT-1372
Created Date:	2026/10/03 20:58:28 (PKT)
Case Directory:	C:\Users\win 11\OneDrive\Desktop\Week8_DigitalForensics_AQT-1372\Autopsy_Case\Week8_DigitalForensics_AQT-1372
Case Type:	Single-user case
Database Name:	C:\Users\win 11\OneDrive\Desktop\Week8_DigitalForensics_AQT-1372\Autopsy_Case\Week8_DigitalForensics_AQT-1372\autopsy.db
Case UUID:	week8_digitalforensics_aqt-1372_20261003_205828

Examiner	
Name:	Mariam Fatima
Phone:	
Email:	
Notes:	AstraQuantum Tech Summer of Cybersecurity 2026 - Week 8 Digital Forensics Practical Assignment

Organization	
Name:	Not Specified
Point of Contact:	
Phone:	
Email:	

Figure 1: Autopsy case created successfully with the assigned case name, case number, examiner name, and case directory.

4. Tools Used

The primary tool used for the investigation was **Autopsy 4.23.1**, an open-source digital forensics platform.

Autopsy was used for:

- Creating and managing the forensic case
- Adding the forensic disk image
- Processing the evidence image
- Examining filesystem artifacts
- Checking deleted files
- Reviewing metadata
- Checking available recent/user activity
- Performing keyword searches
- Investigating individual forensic artifacts

Evidence Source

Digital Corpora – NPS Test Disk Images

Evidence File

ntfs1-gen2.E01

Operating Environment

Windows 11

Time Zone

Asia/Karachi (PKT)

5. Adding and Processing the Forensic Image

The ntfs1-gen2.E01 forensic image was added to the newly created Autopsy case as the primary data source.

During configuration, the time zone was set to **Asia/Karachi (GMT+5:00)** and sector size detection was left on automatic detection.

The image was then processed using the selected Autopsy ingest modules. Processing was allowed to reach 100% before artifact investigation was performed.

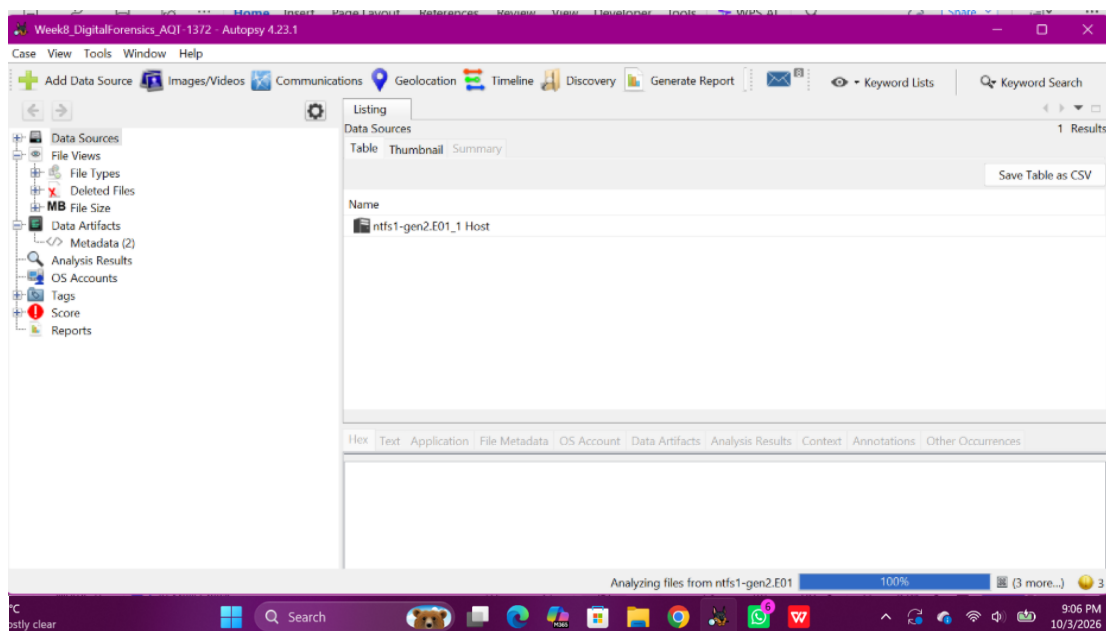


Figure 2: The ntfs1-gen2.E01 forensic image was successfully added to the Autopsy case and processing was completed.

6. Investigation Steps

The following forensic investigation activities were performed:

6.1 Deleted File Analysis

The **File Views** → **Deleted Files** section was examined to identify recoverable deleted files.

No deleted files were identified in the analyzed image.

This result means that Autopsy did not identify recoverable deleted-file artifacts through the available file-system analysis. It does not prove that files were never deleted.

6.2 Recent Activity Analysis

The available Recent Activity/User Activity information was checked after the ingest processing completed.

No Recent Activity/User Activity data was available for the analyzed image in the relevant Autopsy summary.

Therefore, no specific browser or operating-system user activity was attributed to the image based on this artifact category.

6.3 Metadata Analysis

The **Data Artifacts** → **Metadata** section contained six metadata results. Individual artifacts were selected and examined to review their file ownership information, timestamps, descriptions, versions, and source paths.

Two representative metadata artifacts were documented in detail.

6.4 Keyword Search

The **Analysis Results** → **Keyword Hits** section was also examined.

Under **Email Addresses**, Autopsy identified keyword matches including:

bullock@rti.org — 2 hits

xl6@3ov.ke — 1 hit

The bullock@rti.org result was traced to Report02-3.pdf, where the matching text was displayed in the keyword preview.

7. Key Artifact Found – Metadata

7.1 Artifact: 20076517123273.pdf

One of the identified metadata artifacts was:

File Name: 20076517123273.pdf

The metadata extracted by Autopsy included:

Owner: sangli

Version: 1.4

Date Created: 2007-06-05 08:29:54 PKT

Date Modified: 2007-06-05 08:30:13 PKT

Description: Microsoft PowerPoint – 0-A An Overview of NIST and the U.S. Standar...

Source File Path: /img_ntfs1-gen2.E01/Compressed/20076517123273.pdf

Interpretation

The artifact demonstrates that 20076517123273.pdf and its associated metadata were present within the forensic image.

The timestamps provide metadata associated with the file, while the owner field identifies the value recorded in the artifact.

However, the metadata alone does not establish which individual physically created or modified the file. Therefore, the owner value should not be treated as conclusive proof of a person's identity.

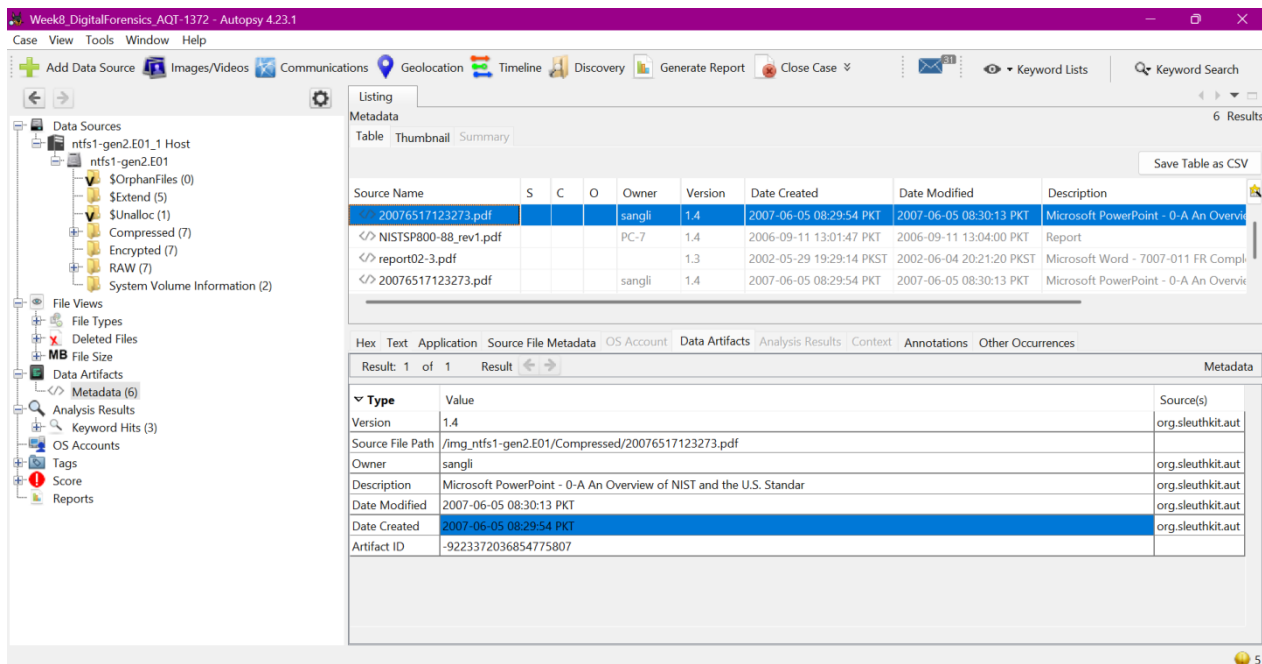


Figure 3: Metadata artifact showing the owner, version, creation and modification timestamps, description, and source file path of 20076517123273.pdf.

8. Additional Artifact – NISTSP800-88_rev1.pdf

A second metadata artifact identified during the investigation was:

File Name: NISTSP800-88_rev1.pdf

The extracted metadata included:

Owner: PC-7

Version: 1.4

Date Created: 2006-09-11 13:01:47 PKT

Date Modified: 2006-09-11 13:04:00 PKT

Description: Report

Source File Path: /img_ntfs1-gen2.E01/Compressed/NISTSP800-88_rev1.pdf

Interpretation

This artifact indicates that NISTSP800-88_rev1.pdf was present in the forensic image and that Autopsy extracted the displayed metadata values.

The PC-7 owner value should not be interpreted as conclusive evidence identifying a particular person. Additional forensic evidence would be required to establish ownership or user attribution.

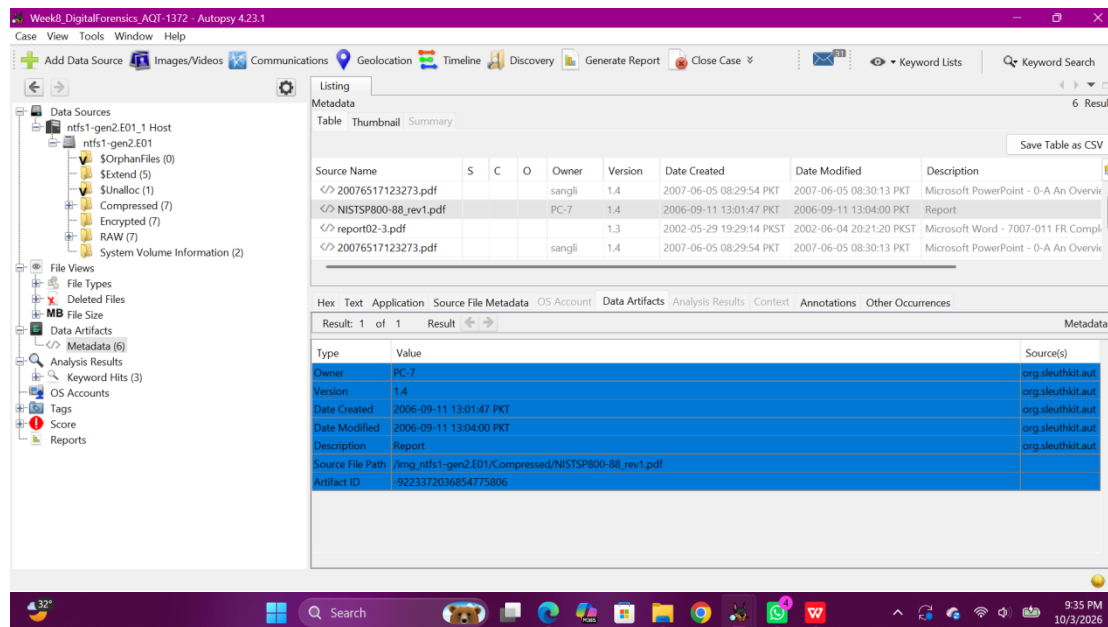


Figure 4: Metadata details for NISTSP800-88_rev1.pdf, including owner, timestamps, description, and source file path.

9. Keyword Search Artifact

Autopsy's keyword search identified email-address-related matches within the forensic image.

The relevant results included:

bullock@rti.org — **2 hits**

xl6@3ov.ke — **1 hit**

The bullock@rti.org result was traced to:

File: Report02-3.pdf

The keyword preview displayed text containing:

(919) 485-5599 or <<bullock@rti.org>>. thank you in

This shows that the email address and an associated telephone number were present within the content of the PDF.

Interpretation

The keyword hit establishes that the email address bullock@rti.org appears within the content of Report02-3.pdf.

However, the presence of an email address inside a document does **not by itself prove that an email was sent or received by the user**. It also does not establish who created or accessed the document. Additional corroborating forensic evidence would be required for those conclusions.

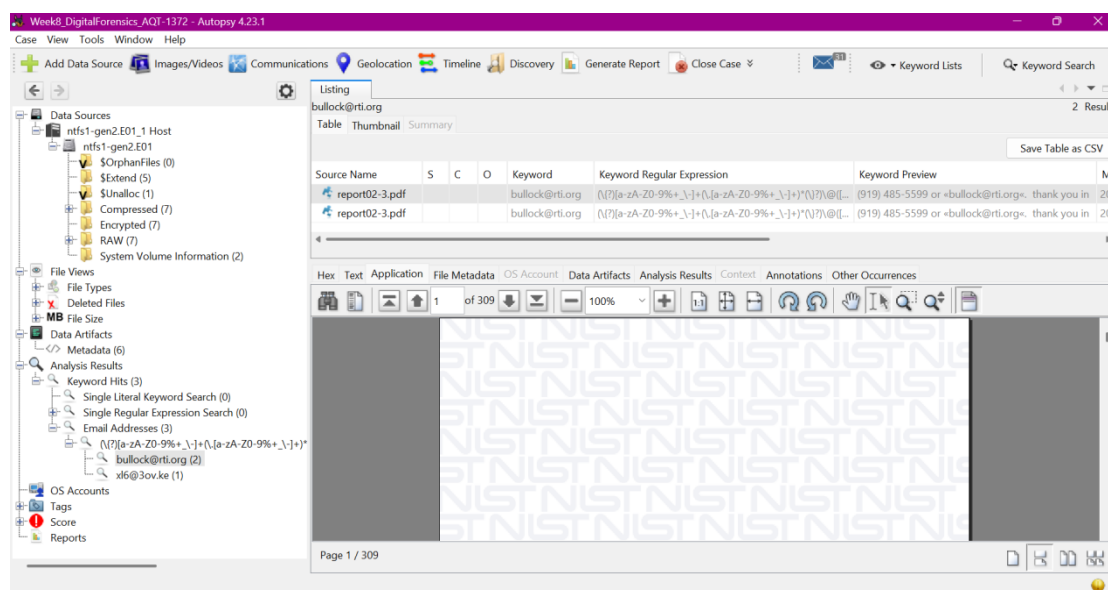


Figure 5: Autopsy keyword search identifying the email address bullock@rti.org within Report02-3.pdf, with the matching content displayed in the keyword preview.

10. Evidence Summary

The main findings from the investigation are summarized below:

Evidence Category	Finding
Forensic Image	ntfs1-gen2.E01 successfully analyzed
Deleted Files	No deleted files identified
Recent Activity	No available Recent Activity/User Activity data
Metadata Artifact 1	20076517123273.pdf – owner recorded as sangli
Metadata Artifact 2	NISTSP800-88_rev1.pdf – owner recorded as PC-7
Keyword Search	bullock@rti.org identified in Report02-3.pdf
Keyword Preview	Email address appeared alongside a telephone number

The artifacts demonstrate that specific files, metadata attributes, and textual information were present within the forensic image.

11. Interpretation

The investigation demonstrates that the selected forensic image contained multiple PDF files with identifiable metadata and searchable textual content.

The metadata artifacts provided file-related information including owner values, timestamps, descriptions, versions, and source paths.

The keyword investigation provided evidence that bullock@rti.org appeared within the contents of Report02-3.pdf.

These findings should be interpreted as evidence of information present within the forensic image rather than definitive proof of individual user behavior. Stronger attribution would require additional corroborating artifacts.

12. Limitations / What the Evidence Does Not Prove

The following limitations apply to the investigation:

The investigation was performed on a publicly available training disk image rather than a live computer system.

No Recent Activity/User Activity data was available in the analyzed image.

No deleted files were identified by Autopsy.

File metadata does not independently prove the identity of the person who created or modified a file.

An owner field does not necessarily establish the actual physical user of a computer.

A keyword hit only establishes that matching text exists within a file.

The presence of an email address in a document does not independently prove that an email was sent or received.

The investigation was limited to artifacts available and successfully processed by Autopsy.

Additional evidence such as system logs, browser history, email records, user-account information, or other corroborating artifacts could be required for stronger attribution.

13. Conclusion

The disk image investigation was successfully completed using Autopsy 4.23.1.

The forensic image ntfs1-gen2.E01 was successfully added to the case and processed. The investigation examined deleted files, recent activity, metadata, and keyword-search results.

The investigation identified metadata associated with 20076517123273.pdf and NISTSP800-88_rev1.pdf. It also identified the email address bullock@rti.org within Report02-3.pdf through keyword searching.

The findings demonstrate the use of Autopsy for examining filesystem artifacts, metadata, timestamps, and searchable content within a forensic disk image. At the same time, the available evidence has limitations and does not independently establish individual user attribution or prove that a communication occurred.